

NORTH SOUTH UNIVERSITY

Department of Electrical and Computer Engineering

EEE 141L / ETE 141L : Electrical Circuit - I Lab



Lab Report Spring 2026

Course Section: 10

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Experiment No. 02

Experiment Name: Kirchhoff's Voltage Law (KVL) and Voltage Divider Rule (VDR) using Series Circuit

Date of Performance: 19/02/2026

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SCORE:	RESUBMIT	REMARKS:	SUBMISSION			
			N	L	SL	R
	UPDATED SCORE		Date:			
	OTHER	PENALTY:	Time:			

□ Experiment Name: Kirchhoff's Voltage Law (KVL) and Voltage Divider Rule (VDR) using series circuit.

□ Objectives: .To learn how to connect a series circuit on a breadboard.

.To observe the voltage distribution across resistors in a series circuit.

.To validate the Voltage Divider Rule (VDR).

.To verify Kirchhoff's Voltage Law (KVL).

□ Theory:

Series Circuit:- A series circuit is an electrical circuit in which components are connected one after another that there is only one path for current to flow, so the same current passes through each component.

A closed series circuit creates a single loop, resulting in equal current through all elements and divided voltage among them.

The total equivalent resistance (R_{eq}) of any number of resistors connected in series is the sum of the individual resistance level of all resistors.

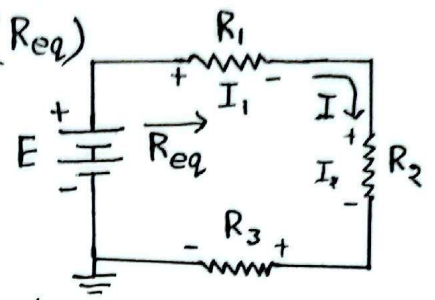


Fig. 1

In Fig. 1, $R_{eq} = R_1 + R_2 + R_3$; $I = \frac{E}{R_{eq}} = I_1 = I_2$

For any number (N) of resistors,

$$R_{eq} = R_1 + R_2 + R_3 + \dots + R_N$$

or,

$$R_{eq} = \sum_{n=1}^N R_n$$

The more resistors we add in series, the greater value of the resistance, no matter what their values are.

Further,

The largest resistor in a series combination will have the most impact on the total resistance.

Kirchhoff's Voltage Law (KVL): Kirchhoff's voltage law specifies that the algebraic sum of all voltages in a closed path or loop is zero.

Expressed mathematically,

$$\sum_{m=1}^M V_m = 0$$

Where Σ represents summation $\oint$ the closed loop, M the number of voltage and V_m the m th voltage.

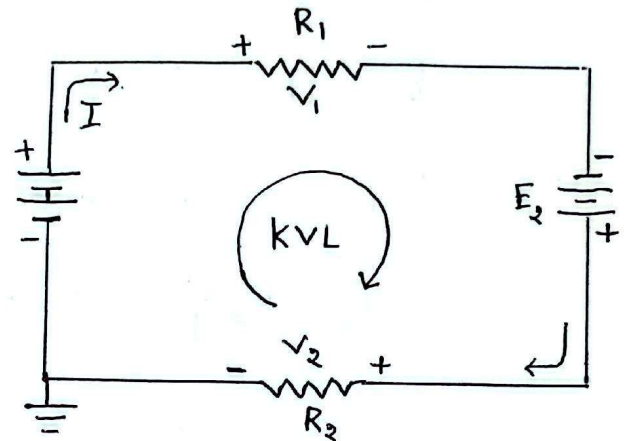


Fig. 2

The applied voltage of a series circuit will equal the sum of the voltage drops of the circuit.
revealing that,

The sum of the potential rises around a closed path will always equal the sum of the potential drops.

can be written as, $\sum_c V_{\text{rises}} = \sum_c V_{\text{drops}}$

If we apply KVL in Fig. 2:

It results,

$$E_1 - V_1 + E_2 - V_2 = 0$$

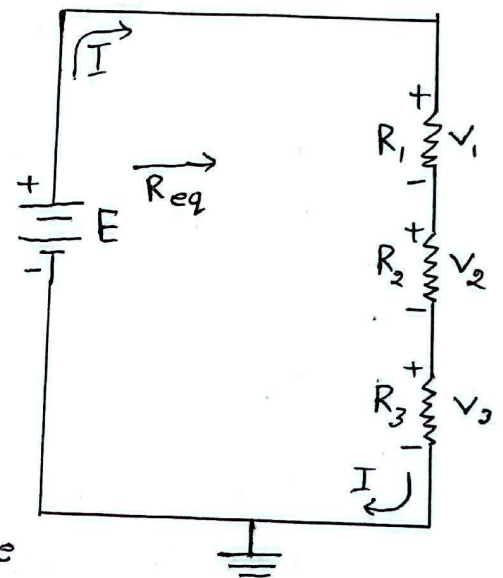
$$\text{or, } E_1 + E_2 = V_1 + V_2$$

Voltage Divider Rule (VDR): The voltage divider rule states that the voltage across a resistor in a series circuit is equal to the value of that resistor times the total applied voltage divided by the total resistance of the series configuration.

Expressed mathematically,

$$V_x = R_x \frac{E}{R_{eq}} = R_x \frac{E}{(R_1 + R_2 + \dots + R_N)}$$

In fig. 3 the source voltage E is divided among the among the resistors.



In a series resistive circuit, the

larger the resistance the more of Fig. 3

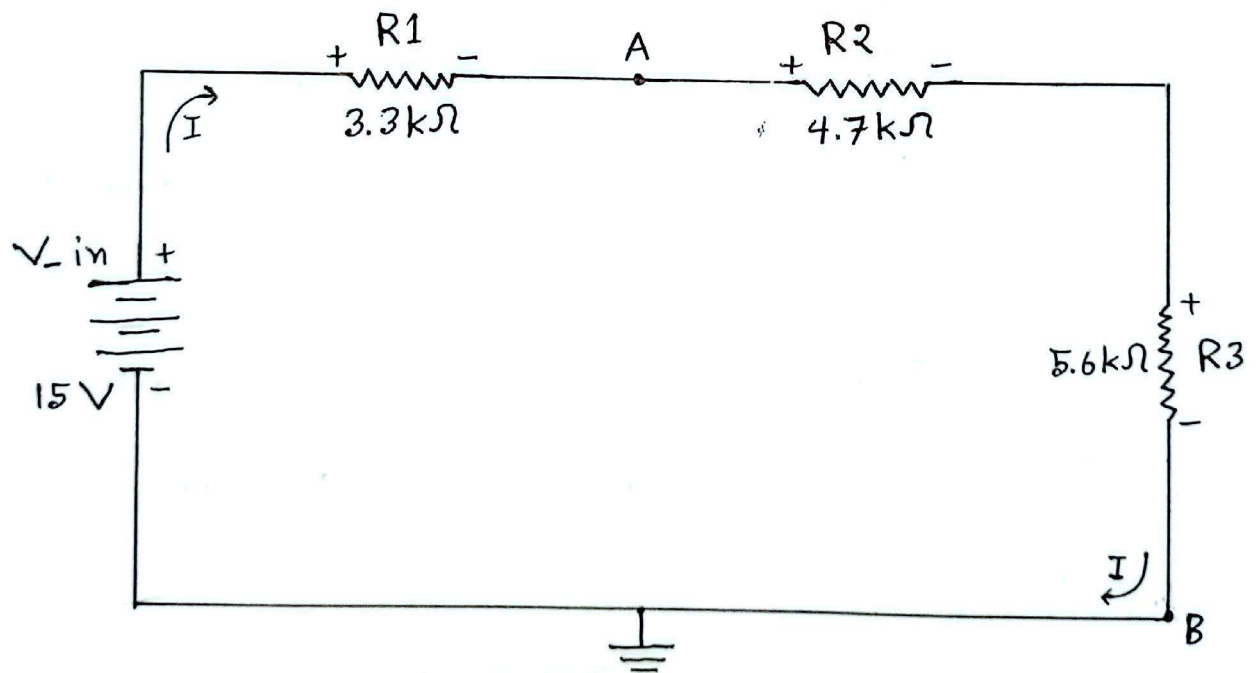
the applied voltage it will capture. Means the divided voltage is directly proportional to the resistance of the resistors. The ratio of the voltages across series resistors will be the same as the ratio of their resistance level.

VDR permits the determination of the voltage across a series resistor without first having to determine the current of the circuit. Ex. $V_2 = R_2 \frac{E}{R_{eq}}$

List of Equipments:

Component Name	Value (or Model Number)	Quantity
Breadboard		1
DC Power Supply (Variable)		1
Resistors	$3.3\text{ k}\Omega$	1
	$4.7\text{ k}\Omega$	1
	$5.6\text{ k}\Omega$	1
Connecting Wires		10
Digital Multimeter (DMM)		1

Circuit Diagram:



Circuit.1

□ Procedure :

(1) First we measure the resistance value of each resistor using DMM and write the practical values in Table 1.

(2) Then we connect three resistors in series and measure the total equivalent resistance using DMM.

(3) We calculate the percentage of error.

$$\text{percentage of error} = \frac{|\text{Practical value} - \text{Theoretical value}|}{\text{Theoretical value}} \times 100\%$$

(4) We connect the voltage source and build the circuit.

(5) Using DMM, we find the potential differences across the source V_{in} and resistors R_1, R_2, R_3 and record the readings in Table 2.

(6) We calculate the sum of potential drops and fill up Table 3.

(7) Then we measure V_{ab} using DMM and calculate V_{ab} using Voltage Division Rule (VDR). Note down values in Table 4.

(8) Now we calculate the percentage of errors and fill the values in Table 3 and 4.

□ Results / Data:

Table 1:

Theoretical Value	Practical Value using DMM	% of Error
3.3 k Ω	3.22 k Ω	2.42%
4.7 k Ω	4.57 k Ω	2.77%
5.6 k Ω	5.45 k Ω	2.68%
$R_{eq} = R_1 + R_2 + R_3 = 13.6 \text{ k}\Omega$	13.24 k Ω	2.65%

Table 2:

	Experimental Readings	Theoretical Values	% of Error
V_{in}	15.09 V	15 V	0.6%
V_{R_1}	3.66 V	3.64 V	0.55%
V_{R_2}	5.2 V	5.184 V	0.31%
V_{R_3}	6.2 V	6.176 V	0.39%

Table 3:

Potential Rise V_{in}	15.09 V	Are the voltage rises and drops Equal?
Sum of Potential Drops $V_{R_1} + V_{R_2} + V_{R_3}$	$(3.66 + 5.2 + 6.2) \text{ V}$ $= 15.06 \text{ V}$	☐ Equal ☒ Almost Equal ☐ Not Equal

Table 4:

	Experimental Readings	Theoretical Values	% of Error
V_{ab}	11.4 V	11.36 V	0.35%

□ Question and Answer:

Q1. State the Voltage Division Rule (VDR).

Answer: The voltage division rule states that the voltage across a resistor in a series circuit is equal to the value of that resistor times the total applied voltage divided by the total resistance of the series configuration and the voltage is directly proportional to the resistance across each component. Can be expressed as, $V_1 = R_1 \frac{E}{R_1 + R_2}$ and $V_2 = R_2 \frac{E}{R_1 + R_2}$. where V_1 and V_2 are the voltage across the individual R_1 and R_2 resistor. R_1, R_2 are their respective resistance and E is the total supplied voltage in the circuit.

In $V_1 = R_1 \frac{E}{R_1 + R_2}$, $\frac{E}{R_1 + R_2} = I_s$ which is the total current flow in a series circuit and its same across all component.

So, $V_1 \propto R_1$ or $V_x \propto R_x$.

The higher resistance the higher voltage across it.

Q2. State the Kirchhoff's Voltage Law (KVL).

Answer: Kirchhoff's voltage law states that the algebraic sum of all voltages around a closed path or loop is zero.

In symbolic form, $\sum_{\text{c}} V = 0$

Where $\sum$ represents summation. c the closed loop and V the potential rise and drops.

The law specifies that the sum of the voltage rises around a closed loop will always equal the sum of the voltage drops.

In symbolic form,

$$\sum_{\text{c}} V_{\text{rises}} = \sum_{\text{c}} V_{\text{drops}}$$

If source voltage (E) provided in a series circuit where V_1 and V_2 voltage drops across R_1 and R_2 resistors then,

$$E - V_1 - V_2 = 0$$

or,

$$E = V_1 + V_2$$

Q3. Showing all the steps, calculate R_{eq} . Compare with the experimental value.

Answer:

$$R_1 = 3.3 \text{ k}\Omega$$

$$R_2 = 4.7 \text{ k}\Omega$$

$$R_3 = 5.6 \text{ k}\Omega$$

R_1 , R_2 and R_3 are in series configuration.

$$\begin{aligned} \therefore R_{eq} &= (\cancel{3.3} R_1 + R_2 + R_3) \\ &= (3.3 + 4.7 + 5.6) \text{ k}\Omega \\ &= 13.6 \text{ k}\Omega \end{aligned}$$

$$\text{Experimental } R_{eq} = 13.24 \text{ k}\Omega$$

$$\begin{aligned} \therefore \text{Error} &= \frac{|E.V. - T.V|}{T.V} \times 100\% \\ &= \frac{|13.24 - 13.6|}{13.6} \times 100\% \\ &= \frac{|-0.36|}{13.6} \times 100\% \\ &= \frac{0.36}{13.6} \times 100\% \\ &= 2.65\% \end{aligned}$$

$\therefore$ Experimental value has 2.65% error compared to the theoretical value of total equivalent resistance (R_{eq}).

Q4. Showing all steps, calculate the theoretical values in Table 2. Compare theoretical values to your experimental values and explain whether your circuit follows KVL or not.

Answer:

Theoretical values:-

$$R_1 = 3.3 \text{ k}\Omega, R_2 = 4.7 \text{ k}\Omega, R_3 = 5.6 \text{ k}\Omega$$

$$R_{eq} = R_1 + R_2 + R_3 = (3.3 + 4.7 + 5.6) \text{ k}\Omega = 13.6 \text{ k}\Omega$$

$$V_{in} = 15 \text{ V}$$

$$\therefore V_{in} = I R_{eq}$$

$$\Rightarrow I = \frac{V_{in}}{R_{eq}} = \frac{15 \text{ V}}{13.6 \text{ k}\Omega} = 1.103 \text{ mA}$$

$$\therefore V_{R_1} = I_{R_1} R_1 = I R_1 = (1.103 \times 3.3) \text{ V} = 3.64 \text{ V}$$

$$V_{R_2} = I_{R_2} R_2 = I R_2 = (1.103 \times 4.7) \text{ V} = 5.184 \text{ V}$$

$$V_{R_3} = I_{R_3} R_3 = I R_3 = (1.103 \times 5.6) \text{ V} = 6.176 \text{ V}$$

Experimental readings:

$$V_{in} = 15.09 \text{ V}$$

$$V_{R_1} = 3.66 \text{ V}$$

$$V_{R_2} = 5.2 \text{ V}$$

$$V_{R_3} = 6.2 \text{ V}$$

Error :

$$F_{\text{or}}, V_{\text{in}} = \frac{|E.R. - T.V. |}{T.V.} \times 100\% = \frac{|15.09 - 15|}{15} \times 100\% = 0.6\%$$

$$F_{\text{or}}, V_{R_1} = \frac{|E.R. - T.V. |}{T.V.} \times 100\% = \frac{|3.66 - 3.64|}{3.64} \times 100\% = 0.55\%$$

$$F_{\text{or}}, V_{R_2} = \frac{|E.R. - T.V. |}{T.V.} \times 100\% = \frac{|5.2 - 5.184|}{5.184} \times 100\% = 0.31\%$$

$$F_{\text{or}}, V_{R_3} = \frac{|E.R. - T.V. |}{T.V.} \times 100\% = \frac{|6.2 - 6.176|}{6.176} \times 100\% = 0.39\%$$

So, Potential rise = 15.09 V

and Potential drops = $V_{R_1} + V_{R_2} + V_{R_3}$

$$= (3.66 + 5.2 + 6.2) \text{ V}$$

$$= 15.06 \text{ V}$$

$$\therefore V_{\text{rise}} \approx V_{\text{drops}} \Rightarrow 15.09 \text{ V} \approx 15.06 \text{ V}$$

The sum of voltage drops is almost equal to the source voltage. Minor differences are due to measurement tolerances.

$\therefore$ Our circuit follows Kirchhoff's Voltage Law (KVL).

Q5. Showing all the calculations, theoretically calculate V_{ab} . Compare with the experimental value and verify the voltage division rule (VDR) at the terminal a-b.

Answer:

Theoretical calculation:

$$R_1 = 3.3k\Omega, R_2 = 4.7k\Omega, R_3 = 5.6k\Omega, R_{eq} = 13.6k\Omega$$

$$V_{in} = 15V$$

$$V_{ab} = R_{ab} \frac{V_{in}}{R_{eq}} = (R_2 + R_3) \frac{V_{in}}{R_{eq}} = \frac{(4.7 + 5.6) \times 15}{13.6} V = 11.36 V$$

Experimental Readings:

$$V_{ab} = 11.4 V$$

$$\text{Error} : \frac{|E.R - T.V|}{T.V} \times 100\% = \frac{|11.4 - 11.36|}{11.36} \times 100\% = 0.35\%$$

$$T.V \approx E.R. \Rightarrow 11.36 V \approx 11.4 V$$

The theoretical values and experimental value matches closely.

$$V_{R_1} = 3.66V \text{ and } V_{ab} = 11.4V$$

$$\therefore V_{R_1} + V_{ab} = (3.6 + 11.4)V = 15.06 \approx V_{in}$$

It verifies the Voltage Division Rule (VDR). Because, observing the values and circuit we can say that, voltage across each resistor and a-b terminal is directly proportional to its respective resistance.

□ Result Analysis and Discussion:

In this experiment, we constructed a series circuit with single voltage source (V_{in}) and three resistors (R_1, R_2, R_3) using breadboard and wires. The experiment was conducted to verify Kirchhoff's Voltage Law (KVL) and Voltage Divider Rule (VDR).

Based on theory, the total resistance of a series circuit should equal the sum of individual resistance.

And the source voltage should divide across the resistors proportionally to their resistance value.

Also according to KVL, the sum of all voltage drops should be equal to the source voltage.

The experimental values of resistance were $R_1 = 3.22 \text{ k}\Omega$, $R_2 = 4.57 \text{ k}\Omega$, $R_3 = 5.45 \text{ k}\Omega$, $R_{eq} = 13.24 \text{ k}\Omega$. Whereas the theoretical values were $R_1 = 3.3 \text{ k}\Omega$, $R_2 = 4.7 \text{ k}\Omega$, $R_3 = 5.6 \text{ k}\Omega$ and $R_{eq} = 13.6 \text{ k}\Omega$.

The percentage error in total resistance is 2.65%.

This small error indicates that the measurement was reasonably accurate.

From the voltage measurement the source voltage was $V_{in} = 15.09V$ and the voltage drops across the resistors were $V_{R_1} = 3.66V$, $V_{R_2} = 5.2V$ and $V_{R_3} = 6.2V$. Since, R_3 has the highest resistance value, it has the highest voltage drop and R_1 has the lowest resistance, has the lowest voltage drop.

The experimental results follows the proportionality between voltage and resistance, which confirms the voltage divider rule (VDR).

Also sum of potential drops ($15.06V$) is very close to the measured source voltage ($15.09V$) with small $0.03V$ difference confirms that Kirchhoff's Voltage Law (KVL) is satisfied.

The measured voltage for a-b terminal $11.4V$ is also close to theoretical value $11.36V$ for the combined resistance of R_2 and R_3 , proves that VDR is satisfied in a-b terminal of the circuit too.

Some percentage error for experimental readings can be because of loose connection of wire, resistor tolerance, internal resistance and the accuracy limitation of the DMM.

□ Conclusion: Overall, the experiment was successful. By measuring and calculating data from this series circuit and comparing it to the theoretical values both the Kirchhoff's Voltage Law (KVL) and Voltage Divider Rule (VDR) were verified experimentally.

□ Reference:

- [1] Fundamentals of Electric Circuits, Charles K Alexander
- [2] Introductory Circuit Analysis, Robert L. Boylestad
- [3] Lab Manual
- [4] Slides from lab class
- [5] Datasheet

□ Attachment:

- [1] Printed copies of Multisim Simulations.
- [2] Signed Datasheet.

Data Collection:

Lab 2:

Group No. 03

Instructor's Signature

Sudipto
19/2/26

Table 1:

Theoretical Value	Practical Value using DMM	% of Error
3.3 k Ω	3.22 k Ω	2.42 %
4.7 k Ω	4.57 k Ω	2.77 %
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$R_T = R_1 + R_2 + R_3 = 13.6 \text{ k}\Omega$	13.24 k Ω	2.65 %

Table 2:

	Experimental readings	Theoretical values	% of Error
V_{in}	15.09 V	15 V	0.6 %
V_{R1}	3.66 V	3.64 V	0.55 %
V_{R2}	5.2 V	5.184 V	0.31 %
V_{R3}	6.20 V	6.176 V	0.39 %

Table 3: use data from table 2

Potential rise V_{in}	15.09 V	Are the voltage rises and drops equal?	
Sum of potential drops $V_{R1} + V_{R2} + V_{R3}$	3.66 + 5.2 + 6.20 = 15.06 V	☐ Equal	☒ Almost Equal ☐ Not Equal

Table 4:

	Experimental readings	Theoretical values	% of Error
V_{ab}	11.40 V	11.36 V	0.35 %

Questions

1. State the voltage division rule (VDR).
2. State the Kirchhoff's voltage law (KVL).
3. Showing all the steps, calculate R_{eq} . Compare with the experimental value.
4. Showing all steps, calculate the theoretical values in Table 2. Compare theoretical values to your experimental values and explain whether your circuit follows KVL or not.
5. Showing all the calculations, theoretically calculate V_{ab} . Compare with the experimental value and verify the voltage division rule at the terminal a-b.

MultiSim Simulation

- Simulate the KVL circuit you did in lab class. Fill the data tables. Verify KVL.